

Working with R

Cheat Sheet

Variable Assignment

```
> a <- 'apple'
> a
[1] 'apple'
```

Getting Help

?mean
Get help for a particular function.

Using Packages

`install.packages('dplyr')`
Download and install a package from CRAN. In RStudio, also: Tools > Install Packages...

`library(dplyr)`
Load a package, making all its functions available to use. This has to be done at the start of every script + session.

`dplyr::select()`
Use a particular function from a package. In this case: use `select` from package `dplyr`.

Objects and Environment

More about an object

`tibble::glimpse(iris)` See a summary of a dataframe.
`head(iris)` See the first 6 rows of a dataframe.
`View(iris)` Browse dataframe in new window.
`typeof(iris)` Find the class an object belongs to.

Set Working Directory

`here::here()`
Guess (and set) the working directory (where inputs are found and outputs are sent).

Use Projects in RStudio to keep your code organized!

The Environment

`ls()` List all variables in the environment.
`rm(x)` Remove x from the environment.
`rm(list = ls())` Remove all variables from the environment.

Vectors

Creating Vectors

<code>c(2, 4, 6)</code>	2 4 6	Join elements into a vector
<code>2:6</code>	2 3 4 5 6	An integer sequence
<code>seq(2, 3, by=0.5)</code>	2.0 2.5 3.0	A complex sequence
<code>rep(1:2, times=2)</code>	1 2 1 2	Repeat a vector
<code>rep(1:2, each=2)</code>	1 1 2 2	Repeat elements of a vector

Working with Vectors

<code>sort(x)</code> Return x sorted.	<code>length(x)</code> N of Elements in x.
<code>table(x)</code> See counts of values.	<code>unique(x)</code> See unique values.

Selecting Vector Elements

By Position

`x[4]` The fourth element.
`x[-4]` All but the fourth.
`x[2:4]` Elements two to four.
`x[-(2:4)]` All except two to four.
`x[c(1, 5)]` Elements one and five.

By Value

`x[x == 8]` Return elements which are equal to 8.
`x[which(x == 8)]` Return position of elements equal to 8.
`x[x < 0]` Return elements less than zero.
`x[x %in% c(1:5)]` Elements in the set {1, 2, 3, 4, 5}.

Named Vectors

`x["apple"]` Element with name "apple".

Types

Converting between common data types

<code>as.logical</code>	TRUE, FALSE, TRUE	Boolean values (TRUE or FALSE)
<code>as.numeric</code>	1, 0, 1	Integer or floating point numbers
<code>as.character</code>	"1", "0", "1"	Character strings. Generally preferred to factors
<code>as.factor</code>	"yes", "no", "yes" levels: "yes", "no"	Character strings with levels. Needed for some statistical models

You can also query whether a vector or dataframe column is of a certain type. These functions return TRUE or FALSE. The following functions ask "Is it a..."

<code>is.numeric(x)</code>	Number?	<code>is.character(x)</code>	String?
<code>is.double(x)</code>	Double?	<code>is.integer(x)</code>	Integer?

Math Functions

Use argument `na.rm = T` if you have NA in a vector, e.g., `sum(x, na.rm = T)`

<code>sum(x)</code>	Sum	<code>round(x, n)</code>	Round to n decimals
<code>mean(x)</code>	Mean	<code>var(x)</code>	The variance
<code>max(x)</code>	Largest element	<code>sd(x)</code>	The standard deviation
<code>min(x)</code>	Smallest element	<code>median(x)</code>	Median

Strings

See also `stringr`

<code>paste(x, y, sep = " ")</code>	Join multiple vectors together.
<code>paste(x, collapse = " ")</code>	Join elements of a vector together.
<code>grep(pattern, x)</code>	Find regular expression matches in x.
<code>gsub(pattern, replace, x)</code>	Replace matches in x with a string.
<code>toupper(x)</code>	Convert to uppercase.
<code>tolower(x)</code>	Convert to lowercase.
<code>nchar(x)</code>	Number of characters in a string.
<code>str_glue("X is {x}")</code>	Return a character vector containing a text and variable values.

Conditionals and Operators

<code>a == b</code>	Are equal	<code>a > b</code>	Greater than	<code>a >= b</code>	Greater than or equal to
<code>a != b</code>	Not equal	<code>a < b</code>	Less than	<code>a <= b</code>	Less than or equal to
<code>is.na(a)</code>	Is missing	<code>is.null(a)</code>	Is null		

You can combine conditions with `&` (and) as well as `|` (or)